

llm-jp/llm-jp-3-13b like 13 Follow LLM-jp 313

Text Generation Transformers Safetensors English Japanese
llama text-generation-inference License: apache-2.0

Train Deploy Use this model

Model card Files xet Community

Downloads last month
1,458



Safetensors

Model size 13.7B params Tensor type BF16 Files info

Inference Providers

Text Generation

This model isn't deployed by any Inference Provider.

Ask for provider support

Model tree for llm-jp/llm-jp-3-13b

Adapters	24 models
Finetunes	1097 models
Merges	1 model
Quantizations	12 models

Collection including llm-jp/llm-jp-3-13b

LLM-jp-3 Pre-trained Models

Collection

Pre-trained models in the LLM-jp-3 model... • 10 items • Updated May 28 • 6

llm-jp-3-13b

This repository provides large language models developed by the [Research and Development Center for Large Language Models](#) at the [National Institute of Informatics](#).

The development was partially supported by [GENIAC](#).

Model Variants

[llm-jp-3-1.8b](#)

[llm-jp-3-1.8b-instruct](#)

[llm-jp-3-3.7b](#)

[llm-jp-3-3.7b-instruct](#)

[llm-jp-3-13b](#)

[llm-jp-3-13b-instruct](#)

[llm-jp-3-172b-beta1](#)

[llm-jp-3-172b-beta1-instruct](#)

Checkpoints format: Hugging Face Transformers

[🔗 Required Libraries and Their Versions](#)

- torch>=2.3.0
- transformers>=4.40.1
- tokenizers>=0.19.1
- accelerate>=0.29.3
- flash-attn>=2.5.8

[🔗 Usage](#)

```
import torch
from transformers import AutoTokenizer, AutoModelForCausalLM
tokenizer = AutoTokenizer.from_pretrained("llm-jp/llm-jp-3-13b")
model = AutoModelForCausalLM.from_pretrained("llm-jp/llm-jp-3-13b", device_map="auto")
```

```

text = "自然言語処理とは何か"
tokenized_input = tokenizer.encode(text, add_special_tokens=False, return_tensors=
with torch.no_grad():
    output = model.generate(
        tokenized_input,
        max_new_tokens=100,
        do_sample=True,
        top_p=0.95,
        temperature=0.7,
        repetition_penalty=1.05,
    )[0]
print(tokenizer.decode(output))

```

🔗 Model Details

Model type: Transformer-based Language Model

- **Total seen tokens:** 2.1T

Params	Layers	Hidden size	Heads	Context length	Embedding parameters	Non-embedding parameters
1.8b	24	2048	16	4096	407,896,064	1,459,718,144
3.7b	28	3072	24	4096	611,844,096	3,171,068,928
13b	40	5120	40	4096	1,019,740,160	12,688,184,320

🔗 Tokenizer

The tokenizer of this model is based on [huggingface/tokenizers](#) Unigram byte-fallback model. The vocabulary entries were converted from `llm-jp-tokenizer v3.0`. Please refer to [README.md](#) of `llm-jp-tokenizer` for details on the vocabulary construction procedure (the pure SentencePiece training does not reproduce our vocabulary).

[🔗 Datasets](#)

[🔗 Pre-training](#)

The models have been pre-trained using a blend of the following datasets.

Language	Dataset	Tokens
Japanese	Wikipedia	2.6B
	Common Crawl	762.8B
	WARP/PDF	237.3B
	WARP/HTML	2.7B
	Kaken	1.8B
English	Wikipedia	4.7B
	Dolma/CC-head	608.5B
	Dolma/C4	181.6B
	Dolma/Reddit	83.1B
	Dolma/PeS2o	62.9B
	Dolma/Gutenberg	5.5B
	Dolma/Wiki	3.9B
Code	The Stack	114.1B
Chinese	Wikipedia	0.8B
Korean	Wikipedia	0.3B

[🔗 Instruction tuning](#)

The models have been fine-tuned on the following datasets.

Language	Dataset	description
Japanese	ichikara-instruction-004-002	A manually constructed instruction dataset
	answer-carefully-002	A manually constructed instruction dataset focusing on LLMs' safety
	ichikara-instruction-format	A small amount of instruction dataset edited from ichikara-instruction, with some constraints on the output format.
	AutoMultiTurnByCalm3-22B	A synthetic instruction dataset.
	random-to-fixed-multiturn-Calm3	A synthetic instruction dataset.
	wizardlm8x22b-logical-math-coding-sft_additional-ja	A synthetic instruction dataset.
	Synthetic-JP-EN-Coding-Dataset-567k	A synthetic instruction dataset. We used sampled one.
English	FLAN	We used sampled one.

[Evaluation](#)

[llm-jp-eval \(v1.3.1\)](#)

We evaluated the models using 100 examples from the dev split.

Model name	average	EL	FA	HE	MC	MR	MT	NLI	QA	RC
llm-jp-3-1.8b	0.3767	0.3725	0.1948	0.2350	0.2500	0.0900	0.7730	0.3080	0.4629	0.7040
llm-jp-3-1.8b-instruct	0.4596	0.4280	0.1987	0.3250	0.3300	0.4200	0.7900	0.3520	0.4698	0.8224
llm-jp-3-3.7b	0.4231	0.3812	0.2440	0.2200	0.1900	0.3600	0.7947	0.3800	0.4688	0.7694
llm-jp-3-3.7b-instruct	0.5188	0.4191	0.2504	0.3400	0.5000	0.5800	0.8166	0.4500	0.4881	0.8247

Model name	average	EL	FA	HE	MC	MR	MT	NLI	QA	RC
<u>instruct</u>										
<u>llm-jp-3-13b</u>	0.5802	0.5570	0.2593	0.4600	0.7000	0.6300	0.8292	0.3460	0.5937	0.8469
<u>llm-jp-3-13b-instruct</u>	0.6168	0.5408	0.2757	0.4950	0.9200	0.7100	0.8317	0.4640	0.4642	0.8500

Japanese MT Bench

We evaluated the models using `gpt-4-0613`. Please see the [codes](#) for details.

Model name	average	coding	extraction	humanities	math	reasoning	roleplay	stem	writing
<u>llm-jp-3-1.8b-instruct</u>	4.93	1.50	4.70	7.80	1.55	2.60	7.80	6.10	7.40
<u>llm-jp-3-3.7b-instruct</u>	5.50	1.95	4.05	8.25	2.25	4.00	8.80	7.25	7.45
<u>llm-jp-3-13b-instruct</u>	6.47	3.15	7.05	9.15	3.75	5.40	8.30	7.50	7.45

Risks and Limitations

The models released here are in the early stages of our research and development and have not been tuned to ensure outputs align with human intent and safety considerations.

Send Questions to

llm-jp(at)nii.ac.jp

License

Model Card Authors

The names are listed in alphabetical order.

Hirokazu Kiyomaru and Takashi Kodama.